



Business Case

A global luxury retail brand is facing a critical challenge: a growing portion of its client base becomes inactive after an initial period of engagement. These clients stop purchasing and gradually disengage from marketing communications, leading to lost revenue and diminished lifetime value.

Reactivating these clients has become a strategic priority for the CRM team. They believe that with the right data-driven approach, it is possible to identify which inactive clients are most likely to return and how best to reach them.

To support this effort, the CRM team has asked you — the Data Scientist — to explore the available data and uncover actionable insights. Your mission is to help the business understand the dynamics of client inactivity and propose a strategy to re-engage the most promising profiles through targeted outreach.

Datasets

- *customers.csv: contains demographic and registration information. The data is current as of today*
- *transactions.csv: contains purchase events between June 1, 2024 and December 31, 2024*
- *campaign_tasks.csv: contains marketing contacts between January 1, 2023 and December 31, 2024*

 **Inactivity Rule:** A client is considered inactive after 120 days since their last purchase

The CRM team has not specified a modeling approach. Instead, they are seeking strategic, data-driven guidance to better understand and address client inactivity. Your role is to explore the data, extract meaningful insights, and propose a solution that supports reactivation efforts. Your task is to explore the data and find an answer to the following business questions:

1. What can you tell about the client population?
2. Can we estimate the probability that an inactive client will return?
3. How do different communication channels (e.g., email, call, SMS, letter) influence the likelihood of reactivation?
4. How should we segment and prioritize inactive clients to support targeted re-engagement campaigns?



Approach Required

Treat this case as a real-world business project. Do not simply answer the questions — instead, follow a classic and structured data science approach, as you would when tackling a new business challenge.

Deliverables: power point presentation, full code (R or Python)